CSC360 Lecture 1

1. How is computer graphics different from image processing?

A1. Computer graphics is mainly about generating and creating images using a computer including animations and certain types of graphics while Image processing deals with accessing and analysing and modifying and understanding existing images consisting of methods like interpolation etc.

2. What are geometric primitives?

A2. Geometric primitives are the basic building blocks of graphics, such as points, lines, curves, circles, rectangles and polygons existing as the basic foundation blocks for any computer graphics of any forms.

3. Describe graphical frameworks in Java.

A3. Java provides graphical frameworks such as AWT, Swing and JavaFX. AWT provides basic GUI and graphics components, Swing provides good GUI components, and JavaFX is used for modern interactive graphical applications.

4. What is the design pattern for creating user interfaces?

A4. The Event Listener Method/Function is used so when the user performs an action, such as clicking or moving the mouse, an event is generated. The listener detects the event and executes the appropriate event-handling code in reaction to the click or hover events as defined in the method.

5. How is static graphics different from interactive graphics?

A5. Static graphics display a fixed image and don't respond to user actions. Interactive graphics respond to events such as clicking, hovering, dragging or keyboard inputs and the graphics can be reactive accordingly.

6. How is a curve connected to calculus?

A6. Calculus helps us understand curves through concepts such as continuity, derivatives, integration, slopes and smoothness. These concepts help derive params like Area, perimeter without extensive infinite calculations.

7. Describe SSH and HTTPS protocols for accessing Git repositories.

A7. SSH i.e. Secure Shell provides secure access using a public/private key pair. HTTPS provides secure access through HTTP with encryption and authentication using credentials or tokens. Both can be used to access Git repositories.

8. Mention tools for SSH operations.

A8. Common SSH tools include ssh, ssh-keygen, ssh-agent, ssh-add and scp. These commands are used to securely connect to a remote server through a secure portal of SSH.

9. What are public and private keys?

A9. A public key can be shared with services such as GitHub, while the private key must be kept secret on your computer. Together they are used for secure authentication and connection to the server.

10. What is a digital image?
A10. A digital image can be represented as a matrix or grid of pixels. Each pixel contains information about the intensity or colour at that particular location. Each pixel representation consists of RGB Value or black and white intensity grey levels.

11. What values does a pixel contain?
A11. In a grayscale image, a pixel generally contains one intensity value. In an RGB image, each pixel contains three values representing Red, Green and Blue forming the colour value.

12. What is the difference between raster and vector graphics?
A12. Raster graphics represent an image using pixels, while vector graphics represent objects using mathematical descriptions such as points, lines and curves. Vector graphics can generally be scaled without losing quality.